TRIGGER LEVEL ANALYSIS DATA PREPARATION THROUGH CO-PROCESSORS

September 12, 2026

AUTHOR(S):
Emilly Raiane Rodrigues

CERN openlab Summer Student Programme

SUPERVISOR(S):
Ioannis Xiotidis and Noah Clarke Hall



PROJECT SPECIFICATION

Trigger decisions made under strict hardware latency constraints can reduce the physics information available for later analysis. Trigger-Level Analysis systems aim to recover part of this information by processing events that would otherwise be discarded, but their implementation requires transmitting substantially more detector data than is normally available outside the hardware trigger.

The planned ATLAS Global Trigger for the HL-LHC includes advanced FPGA resources and AI Engines that may provide additional processing capacity before data transmission. The objective of this project was therefore to investigate whether calorimeter information could be pre-processed directly in the trigger and compressed into physics-level objects, with jets used as the main reconstruction target. The work focused on developing and evaluating lightweight reconstruction algorithms, including machine-learning approaches, capable of transforming the full calorimeter representation into a compact jet list while considering the bandwidth and hardware constraints relevant for a possible Trigger-Level Analysis system.

The project was carried out within the Next-Generation Triggers WP2.1 activities on the use of AI Engines and machine-learning algorithms in the ATLAS Global Trigger.



ABSTRACT

The ATLAS Level-0 Global Trigger at the HL-LHC must process high-granularity calorimeter information under strict bandwidth and latency constraints. This work investigates whether the calorimeter input can be converted directly into a compact list of reconstructed jets, reducing the amount of information that needs to be transmitted while preserving the physics content of the event.

A convolutional baseline model with 708 trainable parameters reaches 77.3% efficiency at 78.0% purity. The reconstruction shows high candidate acceptance, indicating that the main limitation is not finding jet-like structures but distinguishing hard-scatter jets from background activity, particularly at low p_T. Introducing candidate-level self-attention improves the efficiency to approximately 80.5% and the global objectness AUC from 0.934 to 0.950. Control studies show that most of this improvement can be preserved with a substantially smaller convolutional representation, while candidate comparison remains important for low-p_T discrimination.

The dominant remaining limitation is associated with pile-up. When the same reconstruction is trained and evaluated without pile-up, the efficiency increases to 94.2% and the AUC to 0.997, showing that much of the ambiguity comes from the calorimeter information itself under high-pile-up conditions. Finally, the reconstructed jet list contains only 80 values per event, corresponding to approximately 2.1 Gb/s per board, well below the assumed 20 Gb/s bandwidth. The results therefore demonstrate that object-level compression is a promising approach, while FPGA resource usage, latency and the remaining low-p_T ambiguities require further study.

TABLE OF CONTENTS

1 Context 4
- 1.1 Pile-up . . . 4
- 1.2 The trigger and the HL-LHC upgrade . . . 4
- 1.3 Conventions . . . 5

2 Problem and constraints 5
- 2.1 Sample and input representation . . . 5
- 2.2 Required output . . . 6
- 2.3 The bandwidth constraint . . . 7

3 What the calorimeter alone provides 7
- 3.1 Geometric limitation . . . 7

4 Baseline model 8
- 4.1 Output representation . . . 8
- 4.2 Architecture . . . 8
- 4.3 Training . . . 9
- 4.4 Model and output size . . . 10

5 Evaluation 10

6 Baseline results 11
- 6.1 Where the jets are lost . . . 11
- 6.2 Objectness discrimination . . . 11
- 6.3 Reconstruction quality and energy scale . . . 12
- 6.4 Limitation of the objectness score . . . 12

7 Attention-based model 13
- 7.1 Dense prediction stage . . . 13
- 7.2 Candidate pool and comparison . . . 14
- 7.3 Training objective . . . 15
- 7.4 Results . . . 15

8 Control studies 17
- 8.1 Reconstruction without pile-up . . . 18
- 8.2 Architecture controls . . . 18
- 8.3 Bandwidth and FPGA considerations . . . 20

9 Conclusion 21
- 9.1 What limits the reconstruction . . . 21
- 9.2 Architectural choices . . . 21
- 9.3 Bandwidth and remaining work . . . 22

10 Next steps 22

11 References 24

1 Context

Quarks and gluons produced in high-energy proton collisions cannot be observed as free particles. Because of colour confinement they fragment and hadronise into a collimated spray of hadrons travelling approximately along the direction of the original parton. This spray is reconstructed experimentally as a *jet*, whose measured direction and momentum carry information about the quark or gluon from which it originated [11]. Jets are therefore important objects in the identification and selection of many processes at the LHC, including at trigger level.

Reconstructing a jet is more involved than identifying the signature of a single particle. Each of its many hadrons can produce its own shower in the calorimeters, so the jet energy is distributed over several cells and can extend through different calorimeter layers, producing a relatively broad pattern of energy deposits rather than a single localised signature. In the offline reconstruction, calorimeter cells are first grouped into topological clusters, which are then used as inputs to a jet reconstruction algorithm [4, 1]. The output is a set of jets described by quantities such as their energy, momentum and direction.

1.1 Pile-up

In proton-proton collisions at the LHC, multiple interactions can occur within the same bunch crossing. The interaction involving the high-momentum process is referred to as the *hard-scatter interaction*, while the additional interactions are referred to as *pile-up*. An individual calorimeter cell does not identify which interaction produced a given contribution to its measured signal, so the calorimeter signature of a jet is superimposed on a diffuse background of pile-up deposits, which also introduces local fluctuations in the measured energy.

1.2 The trigger and the HL-LHC upgrade

Proton bunches cross at the interaction points of the LHC at a rate of approximately 40 MHz. Recording the full detector information from every bunch crossing is not feasible, so the ATLAS trigger system reduces this rate by selecting events in real time. During Run 3 this selection is performed in two stages [3]: the hardware-based Level-1 trigger processes a reduced set of calorimeter and muon information and reduces the event rate to about 100 kHz, after which the software-based High-Level Trigger applies more detailed reconstruction before events are sent to permanent storage.

The operating conditions will become considerably more challenging at the High-Luminosity LHC, where the number of proton-proton interactions per bunch crossing is expected to reach approximately 200 [6]. This increases detector occupancy and the contribution from pile-up while the bunch-crossing rate remains approximately 40 MHz. The ATLAS trigger and data-acquisition system is therefore being redesigned as part of the Phase-II upgrade [9]. In the planned architecture shown in Fig. 1, a hardware-based Level-0 trigger reduces the 40 MHz input rate to up to 1 MHz, after which a software-based Event Filter reduces the rate to about 10 kHz for permanent storage.

A key element of the Phase-II Level-0 architecture is the *Global Trigger*. It is designed to process calorimeter information with much greater spatial detail than that available to the conventional hardware trigger, including full-granularity calorimeter data [5, 10]. Access to this more detailed information allows more complex reconstruction and selection algorithms to be implemented directly at the hardware trigger level.

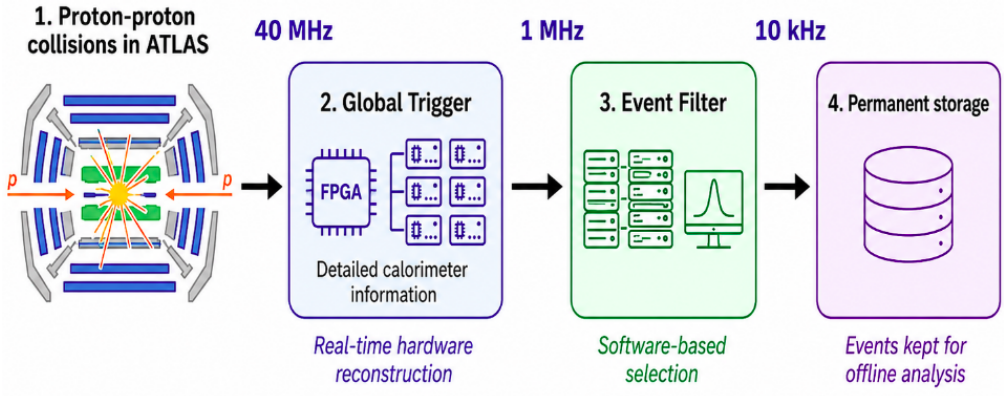


Figure 1: The ATLAS trigger system planned for the HL-LHC. Detector information arrives at 40 MHz and is processed by the Level-0 trigger, which passes selected events at up to 1 MHz to the Event Filter, which reduces the rate to about 10 kHz for storage [9, 10].

1.3 Conventions

Positions in the detector are described using the azimuthal angle ϕ, which specifies the direction around the beam axis, and the pseudorapidity η, which describes the direction relative to it. Values of η close to zero correspond to directions roughly perpendicular to the beam, while larger absolute values correspond to directions closer to it. Calorimeter information can therefore be represented on an η–ϕ plane, obtained by unrolling the cylindrical detector geometry. The angular separation between two positions in this plane is expressed as

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}, \tag{1}$$

where $\Delta\phi$ is taken as the smallest angular separation between the two directions because ϕ is periodic. Momentum transverse to the beam axis is denoted by p_T. Together, p_T, η and ϕ describe the kinematic properties of objects produced in the collisions.

2 Problem and constraints

The Level-0 trigger has access to detailed calorimeter information but must operate under strict constraints on bandwidth, memory and processing time. This creates a data-processing challenge, since the full calorimeter response of each event is much larger than the information ultimately needed for a jet-based trigger decision.

This work addresses the problem with a machine-learning algorithm that reconstructs jets directly from calorimeter images. The model receives a calorimeter image defined in the η–ϕ plane, with multiple channels representing different calorimeter depths, and predicts a list of jets, each described by its transverse momentum p_T, its position (η, ϕ) and a prediction score s. The reconstruction therefore also acts as a form of information compression, and its performance must be considered together with the computational and data-transfer resources it requires.

2.1 Sample and input representation

The dataset consists of simulated gluon-fusion Higgs-boson pair production, with both Higgs bosons decaying into $b\bar{b}$. To reproduce the conditions expected at the HL-LHC, approximately 200 additional proton–proton interactions are included in each bunch crossing. For each simulated

event, the generated particles provide a reference for the jets present in the event, while the simulated calorimeter response provides the detector information used as input to the model.

The reference jet collection is constructed from the generated particles and is therefore independent of the calorimeter reconstruction. These jets are referred to as *truth jets* and provide the targets used to train and evaluate the model. Only truth jets with $p_T > 20$ GeV are retained, since at lower transverse momentum the reference collection becomes increasingly affected by pile-up [2].

The model input is built from the calorimeter response of the same event. Each cell contains an energy measurement together with its position and sampling region. The ATLAS calorimeter has 24 sampling regions, including regions at similar depths that are read out separately because they use different detector technologies; to obtain a simpler and regular input representation, these are grouped by depth into six channels [7]. Each channel is mapped onto a 50×64 grid in the η–ϕ plane, covering $-2.5 < \eta < 2.5$ and the full azimuthal range. Stacking the six grids produces a $50 \times 64 \times 6$ calorimeter image, corresponding to 19 200 input values per event, as illustrated in Fig. 2.

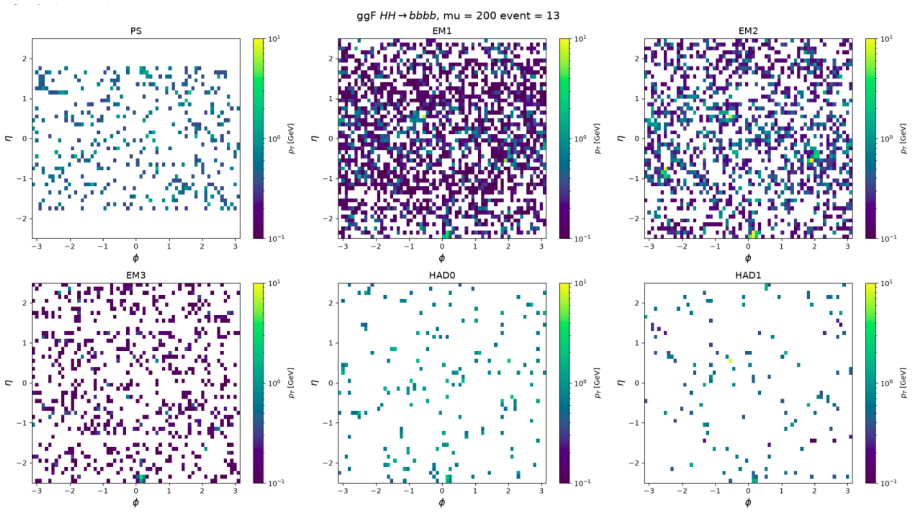


Figure 2: Example of the calorimeter input represented by six depth channels at $\mu = 200$.

2.2 Required output

The model must transform the calorimeter image into a compact list of reconstructed jets $\mathbf{j}_i = (p_{T,i}, \eta_i, \phi_i)$, so that an event is represented by a collection $J = \{\mathbf{j}_1, \ldots, \mathbf{j}_N\}$ where N is not fixed and depends on the event. In the dataset used, most events contain between four and six truth jets, with a median of five and a maximum of eleven, so the output must be able to represent a variable number of jets rather than a fixed-size set of physical objects.

The purpose of this representation is to retain the relevant jet information while reducing the amount of data produced from the calorimeter image. Missing a jet results in the loss of part of the event information, while additional predictions introduce objects that are not associated with the reference collection. Both effects must therefore be taken into account when evaluating the reconstruction.



2.3 The bandwidth constraint

The amount of information that can be processed for each event is limited by the available bandwidth. The Global Trigger processing system operates at the LHC bunch-crossing rate of approximately 40 MHz and uses 49 processing boards [10]. Assuming events are distributed across these boards, each board handles approximately $40 \times 10^6/49 \approx 8.16 \times 10^5$ events/s. For a processing path with approximately 20 Gb/s of available bandwidth, this rate corresponds to a maximum of

$$\frac{20 \times 10^9 \text{ bits/s}}{8.16 \times 10^5 \text{ events/s}} \approx 24\,500 \text{ bits/event}, \tag{2}$$

equivalent at 32-bit precision to approximately 766 values per event. By comparison, the calorimeter input described above contains 19 200 values, or 614 400 bits at the same precision. Transmitting the complete input at the required event rate would therefore require approximately 502 Gb/s per board, about 25 times the available bandwidth.

3 What the calorimeter alone provides

A jet tends to produce a localised region of higher energy in the calorimeter, so local maxima in the energy distribution can be used as simple candidates for its position. To identify these peaks without trainable parameters, the six calorimeter channels are summed cell by cell, the resulting image is averaged over 2×2 blocks, and a cell is kept as a peak when its energy is larger than that of its eight neighbours.

The prominence of each peak is measured relative to the other peaks in the event. If P is the peak energy, P_{median} the median peak energy and $P_{\max}$ the largest one, the visibility is

$$V = \frac{P - P_{median}}{P_{\max} - P_{median}} \times 100\%. \tag{3}$$

The visibility threshold controls how many local maxima are retained as jet candidates. These peaks are matched to truth jets with the Hungarian algorithm, allowing matches only for $\Delta R < 0.4$. The efficiency and purity are then

$$\varepsilon = \frac{N_{\text{matched}}}{N_{\text{truth}}}, \qquad \pi = \frac{N_{\text{matched}}}{N_{\text{pred}}}, \tag{4}$$

where N_{matched} is the number of matched peaks, N_{truth} the number of truth jets and N_{pred} the number of retained peaks.

3.1 Geometric limitation

The visibility threshold controls the balance between the two quantities: lower thresholds retain more peaks and recover more truth jets, while higher thresholds remove more unmatched candidates. In a representative event, all four truth jets are found at $V = 0\%$ but 58 unmatched peaks are retained; at $V = 3\%$ this number falls to 17, at the cost of missing one truth jet.

The jet missed at 3% is less prominent than several unmatched peaks, indicating an intrinsic limit on how clearly some jets can be identified from the calorimeter response alone. The $V > 3\%$ working point is used in the following as a detector-level reference for the reconstruction performance that can reasonably be expected from the neural network.

4 Baseline model

The first architecture developed to produce these outputs is referred to as the *baseline* and provides a simple reference against which later architectures are compared. Its design follows three constraints: a jet is distributed over neighbouring cells and multiple calorimeter depths, so the representation must preserve both local spatial structure and information across channels; the number of jets changes from event to event while the network output must have a fixed size; and the output must remain within the available bandwidth of roughly 766 values per event.

4.1 Output representation

The model produces a fixed set of $K = 20$ candidate slots, each containing

$$\left(\hat{p}_{T,i},\ \hat{\eta}_i,\ \hat{\phi}_i,\ \hat{s}_i\right), \tag{5}$$

where $\hat{s}_i \in [0, 1]$ is the *objectness score*, indicating how likely the candidate is to correspond to a jet. At inference time only candidates above a chosen score threshold are retained, which allows a fixed-size output to represent a variable number of jets. The choice $K = 20$ is large enough to cover the observed jet multiplicities, typically between 4 and 6 and at most 11, while still leaving additional slots to represent nearby calorimeter activity around the truth jets.

4.2 Architecture

The model has two stages, shown in Fig. 3. First, a shared convolutional network processes the calorimeter image and produces three maps over the η–ϕ plane, providing at each position a jet score, an estimate of p_T and a correction to the jet position. The second stage converts these dense maps into a fixed set of candidates: local maxima are identified in the score map, the twenty strongest are retained, and the corresponding p_T and position values are read from the other maps.

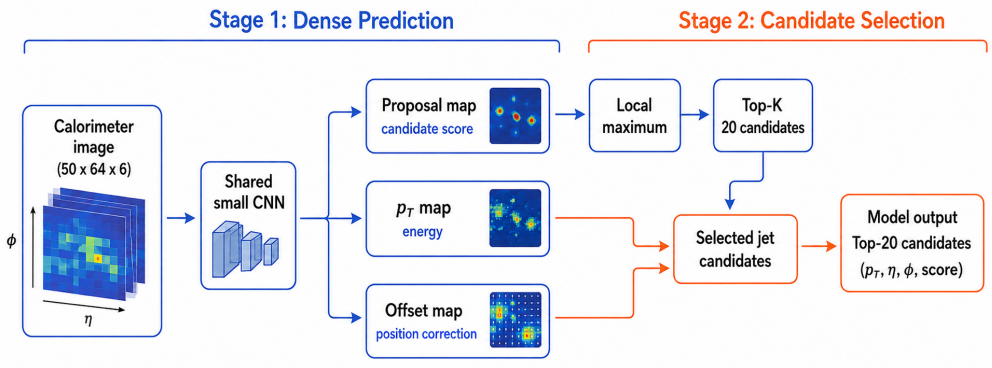


Figure 3: Baseline architecture. The convolutional network produces three maps, from which the twenty strongest jet candidates are selected.

Feature extraction is performed by two shared convolutional layers, summarised in the upper part of Table 1. The first reduces the spatial resolution by a factor of two while combining information from the six calorimeter channels; the second increases the region of the calorimeter seen by each output position without further reducing the resolution. Together they cover an 11×11 region, larger than the typical $R = 0.4$ jet size, allowing the model to compare a

candidate with its surrounding calorimeter activity. Each operation uses padding consistent with the detector geometry, with circular padding in ϕ and zero padding at the boundaries in η.

The shared feature map is then read by three prediction heads, each responsible for one component of the jet candidate. The proposal map uses a sigmoid activation, which constrains the score between 0 and 1; the p_T map uses softplus to keep the predicted transverse momentum positive; and the offset map uses tanh to allow positive and negative corrections in η and ϕ. The proposal and p_T heads are initialised with biases of -1 and -3 as simple starting values without further tuning, while the offset bias is initialised to zero so that the predicted position starts at the centre of the corresponding grid cell.

Table 1: Convolutional layers used for feature extraction (top) and for the prediction heads (bottom).

Layer	Filters	Kernel	Stride	Activation	Output size
Input	–	–	–	–	$50 \times 64 \times 6$
Conv. 1	4	3×3	2	ReLU	$25 \times 32 \times 4$
Conv. 2	4	5×5	1	ReLU	$25 \times 32 \times 4$
Proposal head	1	1×1	1	Sigmoid (bias -1)	$25 \times 32 \times 1$
p_T head	1	1×1	1	Softplus (bias -3)	$25 \times 32 \times 1$
Offset head	2	1×1	1	Tanh (bias 0)	$25 \times 32 \times 2$

The proposal map contains one score for each of the $25 \times 32 = 800$ grid cells. A 3×3 local-maximum operation keeps positions whose score is larger than those of their neighbours, these local maxima are ranked by score, and the $K = 20$ strongest are retained as jet candidates. For each selected position the corresponding p_T and offset values are read from the other two maps. With an average of about 4.7 truth jets per event, twenty slots leave enough room for the true jets as well as additional nearby candidates.

4.3 Training

The p_T and offset maps are trained from the 20 selected candidates using a one-sided Chamfer distance, which for each truth jet finds the prediction with the smallest pairwise distance,

$$\mathcal{L}_{\text{reco}} = \frac{1}{N_{\text{truth}}} \sum_{i=1}^{N_{\text{truth}}} \min_{j \in \{1,\dots,20\}} D_{ij}, \tag{6}$$

where the distance between truth jet i and prediction j is

$$D_{ij} = 1.4\,(p_{T,i} - \hat{p}_{T,j})^2 + 0.6\left[(p_{z,i} - \hat{p}_{z,j})^2 + 2p_{T,i}\hat{p}_{T,j}\left(1 - \cos(\phi_i - \hat{\phi}_j)\right)\right]. \tag{7}$$

The first term measures the difference in transverse momentum, while the second measures the difference in momentum direction and naturally accounts for the periodicity of ϕ. The weight 1.4 gives greater importance to the p_T contribution, leaving 0.6 for the remaining momentum-space term.

Because the Chamfer distance is one-sided, each truth jet is encouraged to have a nearby prediction but extra predictions are not penalised. The proposal score handles this missing part of the objective by assigning low scores to candidates that are unlikely to correspond to real jets. Its loss is therefore trained on the full proposal map rather than only on the 20 selected predictions: all 800 cells contribute, and a cell is labelled positive when it lies within $\Delta R < 0.15$

of a truth jet. Because positive cells account for only about 2% of the map, a standard binary cross-entropy would be dominated by the negative class, so the loss is computed separately for positive and negative cells, each normalised by its own size,

$$\mathcal{L}_{+} = -\frac{1}{N_{+}} \sum_{j:y_j=1} \log(s_j), \qquad \mathcal{L}_{-} = -\frac{1}{N_{-}} \sum_{j:y_j=0} \log(1 - s_j), \tag{8}$$

and both classes are given equal weight, $\mathcal{L}_{\text{proposal}} = 0.5\,\mathcal{L}_{+} + 0.5\,\mathcal{L}_{-}$. The total loss is then

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{reco}} + 0.2\,\mathcal{L}_{\text{proposal}}. \tag{9}$$

The baseline is trained on 76 560 events with 9 570 used for validation, using the Adam optimizer with a learning rate of 3×10^{-4}, a batch size of 32 and early stopping on the validation loss.

4.4 Model and output size

The baseline contains 708 trainable parameters, corresponding to approximately 2.8 kB at 32-bit precision: 220 in the first convolution, 404 in the second, and 84 in the three prediction heads.

The intermediate representations are progressively smaller than the 19 200-value input: the compressed and feature maps contain 3 200 values each, a reduction of a factor of six, and each dense prediction map between 800 and 1 600 values. The feature map alone would still require approximately 83.6 Gb/s per board at 32-bit precision, above the available 20 Gb/s. After candidate selection only $20 \times 4 = 80$ values remain, corresponding to approximately 2.1 Gb/s and reducing the original calorimeter representation by a factor of 240.

5 Evaluation

Model performance is evaluated by matching the predicted jets to the truth jets and measuring how accurately the truth collection is recovered. The same procedure is used for all models.

Truth jets can be close to one another, with the minimum observed separation reaching $\Delta R = 0.380$. A prediction is therefore matched to a truth jet only when their separation satisfies $\Delta R < 0.4$, which reduces the chance of associating it with a neighbouring jet. To avoid multiple predictions being matched to the same truth jet, or one prediction being assigned to several truth jets, the Hungarian algorithm [8] is used to obtain a global one-to-one matching.

Performance is summarised by the efficiency and purity of Eq. 4: efficiency measures the fraction of truth jets that are recovered, while purity measures the fraction of selected predictions that are matched to truth jets. Both depend on the score threshold, so models are compared at the balanced working point where efficiency and purity are equal.

The reconstruction is also evaluated with the AUC, which measures how well the objectness score separates predictions associated with truth jets from unmatched predictions. After the one-to-one matching, matched predictions are treated as signal and unmatched predictions as background, the objectness threshold is varied to construct the ROC curve, and the AUC is the area under it, with values closer to 1 indicating better separation. Unlike efficiency and purity, the AUC uses the full range of thresholds and does not depend on the final working point.

6 Baseline results

At the balanced working point, reached with an objectness threshold of 0.96, the baseline achieves an efficiency of 77.3%, a purity of 78.0% and a global AUC of 0.934. These and all subsequent numbers are quoted on the validation sample, and a comparison of every model studied in this work is collected in Table 3.

6.1 Where the jets are lost

A single efficiency value shows how many truth jets are recovered, but not where the losses occur. As described in Sec. 4, the model reconstructs each jet in two steps: a nearby candidate must first be produced, and this candidate must then pass the objectness threshold. The efficiency can therefore be separated into *acceptance*, the fraction of truth jets with a candidate before the score cut, and *selection*, the fraction of those candidates that survive the cut. Their dependence on truth-jet p_T is shown, together with the attention model, in Fig. 6.

At low p_T, most of the remaining loss comes from the objectness selection rather than from the absence of a nearby candidate. What fails is the ranking, since the threshold needed to control the fake rate removes most of the low-energy candidates along with the fakes. As p_T increases this ambiguity decreases, and above 70 GeV the efficiency exceeds 97%, indicating that sufficiently energetic jets are reconstructed reliably.

This raises the question of whether the lower efficiency at small p_T is mainly a limitation of the model or of the calorimeter information itself. Comparing the baseline with the detector-level reference of Sec. 3.1, both fall at low p_T, indicating that part of the inefficiency in this region comes from the limited visibility of low-energy jets in the calorimeter. Above 70 GeV the model approaches the reference, while the largest gap appears at intermediate p_T, where the calorimeter still contains a visible peak but the model does not always recover it: in the 30–35 GeV bin the reference reaches about 68% while the model reaches 41%, and even in the lowest bin the model reaches only 13% compared with about 43% for the reference. Some visible low-p_T jets are therefore still being lost by the model. The corresponding curves are shown together with the attention model in Fig. 8.

6.2 Objectness discrimination

Since much of the loss at low p_T occurs during the objectness selection, it is useful to determine whether this is caused by the chosen threshold or by limited separation between real and false candidates. The AUC, evaluated independently in each p_T bin, measures the discrimination learned by the proposal map without depending on the selected working point.

The separation improves with p_T. In the lowest bin the AUC is only about 0.68, showing that low-p_T candidates are difficult to distinguish from unmatched candidates using the learned score, and it then rises rapidly and approaches unity above about 70 GeV (Fig. 7). This follows the same trend as the efficiency and indicates that the loss at low p_T is not simply a consequence of the chosen threshold: the candidate signatures themselves become progressively easier to distinguish as the jet energy increases.

The objectness head has therefore largely learned to measure energy rather than to recognise a jet, which follows from its design, since it is a 1×1 convolution reading one cell of the compressed grid with no way of comparing a candidate against the others in the same event. Separating a real 25 GeV jet from a 25 GeV pile-up fluctuation requires exactly that comparison.

6.3 Reconstruction quality and energy scale

The previous results describe whether a jet is found. For the matched jets, the next question is how accurately their properties are reconstructed.

The jet position is reconstructed with high accuracy: the correlations are 0.999 for η and 0.973 for ϕ, with median absolute errors of 0.029 and 0.022 respectively, well below the size of a cell in the compressed grid. The offset prediction therefore successfully recovers sub-cell spatial information. The transverse momentum is less precise; although the correlation remains high at 0.921, the distribution is visibly broader, with a median absolute error of 11.4 GeV. The baseline thus preserves the jet direction more accurately than its transverse momentum, which is also visible in the residual distributions, where the angular residuals remain narrow around zero while the p_T residual is broader and asymmetric.

The weakest correlation is found for the number of reconstructed jets per event, with a value of 0.543. Unlike the jet coordinates and p_T, the final multiplicity depends on which candidates pass the objectness threshold, and it therefore reflects both the reconstruction and the candidate-selection performance rather than a single regression output.

The asymmetry of the p_T residual is a consequence of the energy scale. The median response $p_T^{\text{pred}}/p_T^{\text{truth}}$ is overestimated at low p_T, reaching about 1.6 in the lowest bin, gradually approaches unity as the jet energy increases, crosses unity around 50 GeV and becomes slightly underestimated at high p_T.

6.4 Limitation of the objectness score

A large fraction of the inefficiency, particularly at low p_T, therefore appears after a nearby candidate has already been formed, which points to the objectness score as the main remaining limitation of the baseline. To examine whether the rejected truth-associated candidates were still competitive within their own events, the candidates were ranked by objectness before applying the final threshold. Among the 8 209 truth jets that had a matched candidate but were rejected by the 0.96 cut, 1 971, or about 24%, were still ranked within the truth-jet multiplicity of their event: in an event with five truth jets, for example, a rejected candidate could still appear among the five highest-scoring candidates.

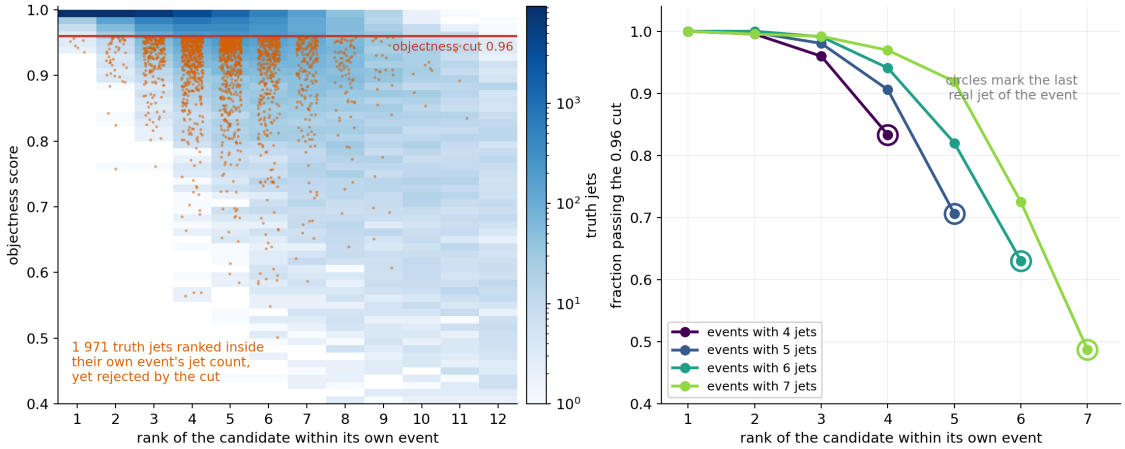


Figure 4: Objectness score as a function of the candidate rank within each event. The right panel shows the fraction of matched candidates passing the 0.96 threshold for different truth-jet multiplicities.

Some truth-jet candidates therefore remain important relative to the other candidates in the

same event while receiving an absolute score below the final threshold, and the effect is stronger towards the end of the ranking: the last matched candidate passes the cut in about 83% of events with four truth jets, decreasing to about 49% for events with seven truth jets.

This behaviour exposes a limitation of the baseline architecture. The proposal score is produced from the local features at each position before the candidates are compared with one another, so the final selection uses a score that does not explicitly contain information about the relative importance of a candidate within its event. This suggests that an event-level comparison could provide useful information for improving the candidate selection.

7 Attention-based model

The baseline study showed that candidate scoring remains an important limitation of the reconstruction: each proposal score is obtained independently from the local calorimeter information around that position, without considering the other candidates in the same event, so a truth-associated candidate can remain competitive within its event and still fall below the single global objectness threshold. This motivates introducing a candidate-comparison stage, so that the final score can depend not only on the local response of a candidate but also on its relation to the other candidates.

Two changes are made. First, the local convolutional representation is made wider, so that more information about the calorimeter response is retained before candidate selection; this does not increase the spatial receptive field, but gives the network a richer learned description of the same local region. Second, the selected candidates are compared with each other through self-attention before their final scores are assigned. The complete chain is shown in Fig. 5.

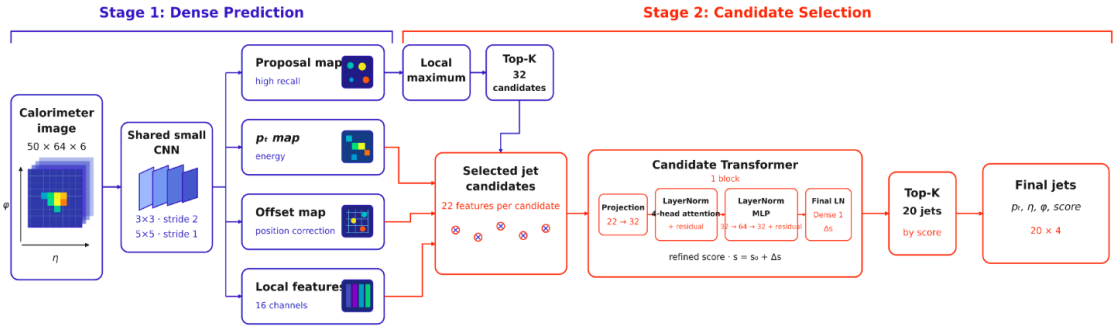


Figure 5: Architecture of the Attention reference model. Local calorimeter information is first converted into proposal, p_T and position predictions. The 32 strongest local proposal maxima form a candidate pool. Their representations are compared by self-attention, which modifies the candidate score before the final twenty jets are selected.

7.1 Dense prediction stage

The first two convolutions follow the same spatial structure as the baseline but use a wider representation. The p_T and offset predictions are obtained from the shared feature map, while candidate scoring has an additional 3×3 convolution with 16 channels producing the *proposal features*, which are then converted into a single proposal map. The configuration is summarised in Table 2.

The initialisations follow from the statistics of the sample. The proposal map contains 800 cells and, although there are on average about five truth jets per event, the $\Delta R < 0.15$ labelling

Table 2: Configuration of the dense prediction stage of the Attention reference model.

Map	Kernel	Channels	Stride	Activation	Initialization	Output
Compressed map	3×3	32	2	ReLU	Default	$25 \times 32 \times 32$
Feature map	5×5	32	1	ReLU	Default	$25 \times 32 \times 32$
Proposal features	3×3	16	1	ReLU	Default	$25 \times 32 \times 16$
Proposal map	1×1	1	1	Linear	Bias $= -4.56$	$25 \times 32 \times 1$
p_T map	1×1	1	1	Softplus	Bias $= 70$	$25 \times 32 \times 1$
Offset map	3×3	2	1	Tanh	Kernel, bias $= 0$	$25 \times 32 \times 2$

region can mark more than one cell as positive for each jet, giving approximately 8.28 positive cells per event and a positive fraction $P = 8.28/800 = 0.01035$. The initial proposal bias is therefore $b_{\text{proposal}} = \log[P/(1-P)] \simeq -4.56$. The p_T bias is initialised to 70 GeV, chosen from the mean truth-jet transverse momentum in the training sample of approximately 69 GeV; since Softplus(70) $\simeq 70$, the model initially predicts a p_T close to the typical jet energy rather than starting near zero. The offset map is initialised at zero, so the initial reconstructed position corresponds to the centre of the compressed grid cell.

7.2 Candidate pool and comparison

Candidate jet centres are identified by searching for local maxima within each 3×3 neighbourhood of the proposal map, after which the 32 positions with the largest proposal logits are retained as the candidate pool. For each candidate the model constructs the feature vector

$$\mathbf{x}_{candidate,i} = \left[\mathbf{f}_i^{\text{prop}}, \log(1+\hat{p}_{T,i}), \Delta\eta_i, \Delta\phi_i, \eta_i^{\text{cell}}, \sin\phi_i^{\text{cell}}, \cos\phi_i^{\text{cell}}\right], \tag{10}$$

where $\mathbf{f}_i^{\text{prop}}$ are the 16 proposal features evaluated at the candidate position, providing a learned description of the local calorimeter response used by the scoring branch. The logarithm of p_T reduces its numerical range, while the offsets and cell coordinates provide the reconstructed spatial information. The 16 proposal features together with the remaining six quantities give 22 values per candidate, so the complete representation is $X_{candidate} \in \mathbb{R}^{32\times22}$. The original proposal logit is not included explicitly; it is kept separately and used as the starting point for the refined score.

Each 22-dimensional candidate vector is projected to 32 dimensions and the 32 candidates are then processed together by one self-attention block with four attention heads, a key dimension of 8 and an MLP of $32 \to 64 \to 32$, with residual connections and layer normalisation around both the attention operation and the MLP. Self-attention allows the representation of each candidate to depend on the other candidates in the same event.

The resulting representation is used only to modify the candidate score. The original proposal logit ℓ_i is retained and the attention block predicts a correction $\Delta\ell_i$, so that $\tilde{\ell}_i = \ell_i + \Delta\ell_i$ and the objectness score is $\hat{s}_i = \sigma(\tilde{\ell}_i)$. The layer producing $\Delta\ell_i$ is initialised with zero weights and zero bias, so that at the beginning of training the refined logit equals the original proposal logit and the model then learns whether each candidate should be promoted or suppressed after comparison with the rest of the event. Finally, the 20 candidates with the largest refined logits are retained; the attention block does not modify their reconstructed p_T, η or ϕ.

7.3 Training objective

The reconstruction loss remains the one-sided Chamfer loss used for the baseline, and two objectness terms supervise the initial proposal map and the refined candidate scores. The proposal loss is evaluated over the complete 25×32 map, with a cell labelled positive when it lies within $\Delta R < 0.15$ of a truth jet and positive and negative cells normalised separately and combined with equal weight; since the map contains logits, binary cross-entropy is evaluated directly from these values.

The candidate loss is evaluated over the pool of 32 candidates. For each truth jet the nearest valid candidate is labelled positive if it lies within $\Delta R < 0.20$, and all remaining valid candidates are labelled negative. Denoting the positive and negative sets in an event by $\mathcal{P}$ and $\mathcal{N}$,

$$\mathcal{L}_{\text{candidate}} = \frac{1}{2}\left(-\frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}}\log\sigma(\tilde{\ell}_j)\right) + \frac{1}{2}\left(-\frac{1}{|\mathcal{N}|}\sum_{j\in\mathcal{N}}\log\left[1-\sigma(\tilde{\ell}_j)\right]\right), \tag{11}$$

so that positive and negative contributions are balanced within each event. The complete objective is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{reco}} + 0.2\,\mathcal{L}_{\text{proposal}} + 0.5\,\mathcal{L}_{\text{candidate}}$, and training uses the Adam optimizer with a learning rate of 3×10^{-4}.

7.4 Results

The Attention reference is evaluated on the same validation sample and with the same matching procedure used for the baseline; the results are included in Table 3.

7.4.1 Where the improvement appears

The attention architecture increases the reconstruction efficiency from 77.3% to 80.5% and the purity from 78.0% to 80.7%. The acceptance before the score requirement changes only from 95.3% to 95.9%, whereas the fraction of accepted candidates surviving the final selection increases from 81.1% to 83.9%, confirming that the gain comes from candidate scoring rather than from candidate formation. The p_T-dependence of both quantities is shown in Fig. 6.

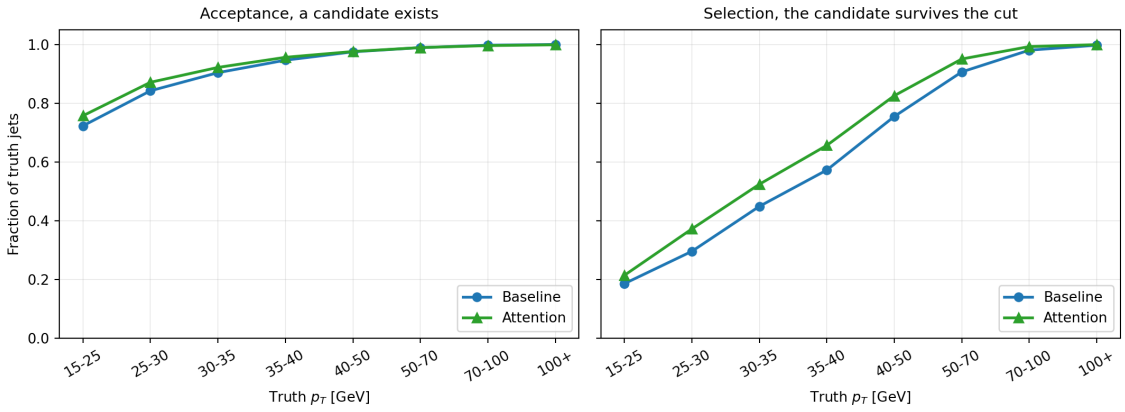


Figure 6: Acceptance and selection in bins of truth p_T for the baseline and Attention reference models.

Truth-matched candidates are concentrated at high refined objectness scores while most unmatched candidates appear at lower values, so the working point near 0.64 removes many unmatched candidates while retaining most truth-associated ones. This improved separation

is quantified by the AUC in Fig. 7: the global value increases from 0.934 to 0.950, with the largest gains at low and intermediate p_T, rising from 0.677 to 0.717 in the 15–25 GeV bin. This improvement cannot be attributed to self-attention alone, however, since the Attention reference also uses a wider convolutional representation and a dedicated proposal-feature branch.

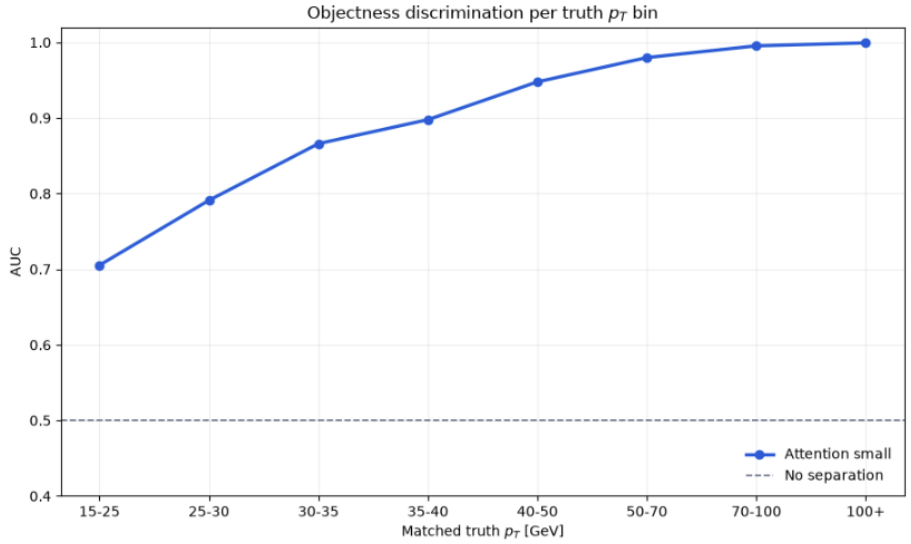


Figure 7: AUC of the objectness score computed within bins of truth p_T for the baseline and Attention reference models, each against the same complete background sample.

Compared with the detector-level visibility reference of Sec. 3.1, shown in Fig. 8, the Attention reference reduces part of the gap left by the baseline, particularly in the intermediate p_T region. The improvement is smaller in the lowest momentum bin, where the reference itself already shows a substantial loss in efficiency.

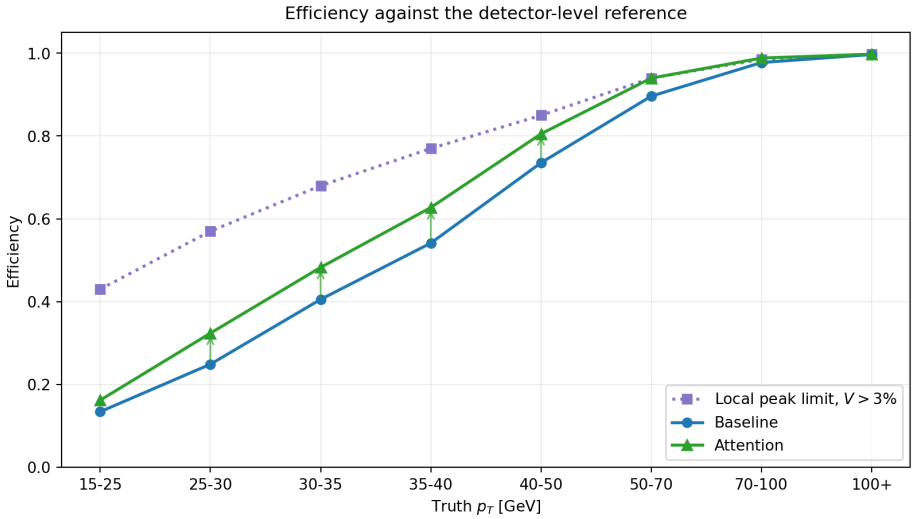


Figure 8: Efficiency of the baseline and Attention reference models compared with the detector-level visibility reference at $V > 3\%$.

7.4.2 Reconstructed jet properties

The attention block modifies only the candidate score, while p_T, η and ϕ are reconstructed by the dense prediction stage. Since the Attention reference also uses a wider convolutional

representation, changes in these quantities cannot be attributed to self-attention alone. The spatial reconstruction remains similar, with the median ΔR changing from 0.044 to 0.046 and the median p_T response from 0.969 to 0.966, while the mean absolute p_T error decreases from 17.3 to 16.0 GeV. The largest difference appears at low p_T, where the upward bias of the response is reduced. These changes may result from the wider convolutional representation and from the different candidates retained after score refinement; the direct role of attention remains candidate scoring rather than the reconstruction of jet energy or position.

7.4.3 Score ranking and event-level behaviour

The original motivation for attention came from the observation that some truth-matched candidates were already highly ranked within their event but still failed the global objectness requirement. Repeating the same diagnostic after candidate comparison, the number of such jets decreases from 1 971 in the baseline to 1 577, corresponding to a reduction from approximately 24.0% to 22.5% of the truth jets lost at the score requirement. Attention therefore reduces, but does not remove, the limitation that motivated candidate comparison.

Because self-attention introduces information from the other candidates in the event, its behaviour can also depend on jet multiplicity, as shown in Fig. 9. For events with 3, 4 and 5 truth jets, the survival probability of the last truth-associated candidate increases by approximately 7, 10 and 7 percentage points, but the trend reverses for larger multiplicities, decreasing by about 3 points for six jets and 17 points for seven jets. The Attention model also predicts a narrower multiplicity range: low-multiplicity events tend to receive more predictions and high-multiplicity events fewer, pulling the prediction towards the typical multiplicity of the sample. This is consistent with the decrease in multiplicity correlation from 0.543 to 0.512, despite the improvements in efficiency, purity and AUC.

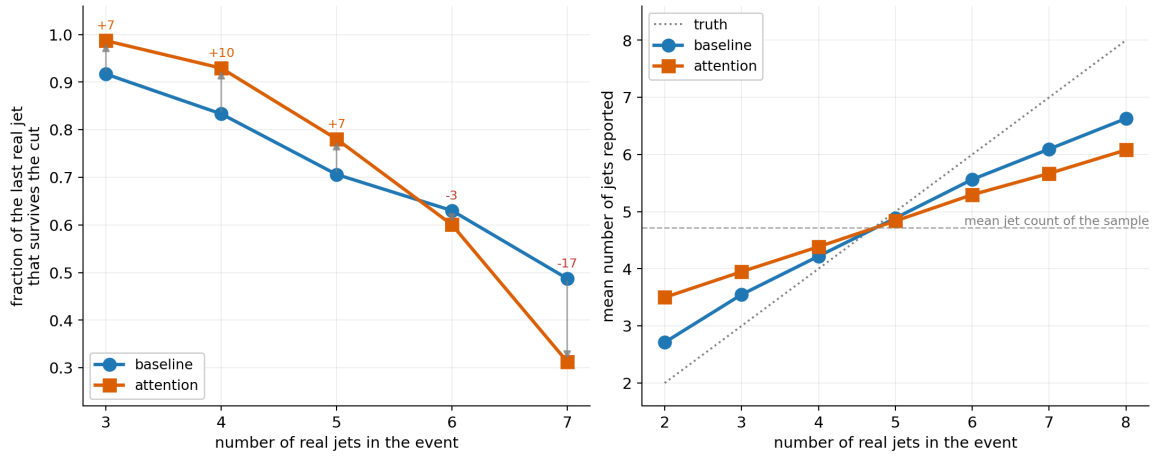


Figure 9: Dependence of the final selection on event multiplicity. Left, fraction of events in which the lowest-ranked truth-associated candidate survives the objectness requirement. Right, mean number of jets reported by the model as a function of truth multiplicity.

8 Control studies

The results so far show that the remaining inefficiency is concentrated at low p_T, where pile-up makes the calorimeter response more difficult to interpret. Two control studies are used to separate the contributions of the data and of the architecture: the same reconstruction is

first retrained on a sample without pile-up, and reduced versions of the architecture are then compared to identify which of its components are responsible for the improvement. Table 3 summarises every model discussed in this work.

Table 3: Validation performance of all models studied in this work. The $\mu = 0$ result is a control measurement and does not represent HL-LHC operating conditions.

Metric	**Baseline**	**Attention V0**	**Attention V1**	**Attention V2**	**V0, $\mu = 0$**
Trainable parameters	708	42 021	9 639	1 155	42 021
Objectness cut	0.960	0.645	0.645	0.640	0.760
Efficiency	77.3%	80.5%	80.3%	79.5%	94.2%
Purity	78.0%	80.7%	80.5%	79.6%	94.3%
Acceptance	95.3%	95.9%	95.4%	95.2%	99.0%
Selection	81.1%	83.9%	84.2%	83.5%	95.2%
Global AUC	0.934	0.950	0.950	0.946	0.997
Median ΔR	0.044	0.046	0.049	0.043	0.040
Median p_T response	0.969	0.966	0.958	0.963	0.990
p_T MAE [GeV]	17.3	16.0	17.4	17.6	12.1

8.1 Reconstruction without pile-up

Pile-up has so far been studied indirectly through the local-peak reference. A direct test is to retrain the same Attention model on a sample without pile-up, keeping the training and evaluation procedure unchanged. This is a control measurement rather than a proposed configuration, since the HL-LHC will run at high pile-up and a $\mu = 0$ model is not deployable. Its value is that it isolates one variable: if the same model performs far better without pile-up, then the architecture is adequate and pile-up ambiguity is the dominant limitation; if it does not, the limitation lies in the model. The sample contains 74 800 training and 9 350 validation events with 43 989 truth jets, so the comparison is between samples of the same process rather than between the same events.

Removing pile-up produces a much larger change than the transition from the baseline to the Attention model at $\mu = 200$. Efficiency increases from 80.5% to 94.2% and purity reaches 94.3%, with the largest change in selection, which rises from 83.9% to 95.2%. Pile-up therefore strongly affects the ability to distinguish which calorimeter responses correspond to the hard-scatter jets.

The same effect is visible in candidate discrimination. The global AUC increases from 0.950 to 0.997, and from 0.717 to 0.977 in the 15–25 GeV bin, so pile-up is a major source of ambiguity in candidate classification. Removing it does not eliminate all losses, however: in the lowest bin the efficiency remains 62.2% despite a 95.4% acceptance, indicating that some nearby candidates are still difficult to select correctly, which can arise from overlapping showers, distributed energy deposits and finite calorimeter granularity. The $\mu = 0$ model also exceeds the local-peak reference in part of the p_T range, confirming that this reference describes task difficulty rather than a strict upper limit.

8.2 Architecture controls

The pile-up study showed that a large part of the reconstruction difficulty comes from the input itself. A separate question is whether the improvement of the Attention reference requires its full 42 021-parameter architecture, since Attention V0 changes both the local candidate

representation and the candidate-comparison stage. Two reduced models are therefore used to study these components separately. Attention V1 keeps the four-channel CNN of the baseline and only one dedicated proposal feature, but retains the full attention block of the reference model; Attention V2 uses the same local representation but also reduces the attention stage to an eight-dimensional token with a single attention head. The two controls therefore receive exactly the same seven candidate features and differ mainly in the capacity of the candidate-comparison stage. Table 4 summarises the configurations.

Table 4: Main dimensions of the baseline, Attention reference and architecture controls.

Property	**Baseline**	**Attention V0**	**Attention V1**	**Attention V2**
Compressed-map channels	4	32	4	4
Feature-map channels	4	32	4	4
Proposal features	–	16	1	1
Candidate features	–	22	7	7
Candidate pool	–	32	32	32
Final candidates	20	20	20	20
Token dimension	–	32	32	8
Attention heads	–	4	4	1
Key dimension	–	8	8	4
MLP hidden dimension	–	64	64	8
Trainable parameters	708	42 021	9 639	1 155

In all attention models the candidate representation includes the same six reconstructed quantities, $\log(1+\hat{p}_T)$, the two offsets, the cell η, and $\sin\phi$ and $\cos\phi$; the difference is the local calorimeter information added to them, 16 proposal features in V0 against a single one in V1 and V2. One proposal feature is retained so that the attention stage still receives local information for jet discrimination, since the other inputs mainly describe the reconstructed jet energy and position, while the proposal feature can encode differences in the local calorimeter response between real jets and background activity.

Attention V1 reaches almost the same global performance as Attention V0 despite using only one proposal feature and a four-channel convolutional representation, with an efficiency of 80.3% against 80.5% and the same global AUC of 0.950. Reducing the attention block as well produces a loss: Attention V2 reaches 79.5% efficiency and an AUC of 0.946. Since the two controls use the same local representation and differ only in the size of the attention block, the larger candidate-comparison stage recovers part of the lost performance. The remaining difference between V0 and V1 is instead associated with the local reconstruction stage, where candidate acceptance decreases from 95.9% to 95.4% and the p_T mean absolute error increases from 16.0 to 17.4 GeV. The larger convolutional representation therefore remains useful for candidate formation and energy reconstruction, even when the scoring performance is largely recovered.

The effect of the attention size is most visible at low p_T, as shown in Fig. 10. In the 15–25 GeV bin the AUC increases from 0.698 for Attention V2 to 0.719 when the full attention block is restored, essentially matching the 0.717 of Attention V0, with corresponding efficiencies of 14.8%, 16.3% and 16.2%. The full attention block is therefore particularly useful for separating weak jet candidates from background activity. All three models nevertheless remain well below the local-peak reference at low p_T, showing that candidate scoring is only one part of the remaining low-energy inefficiency.

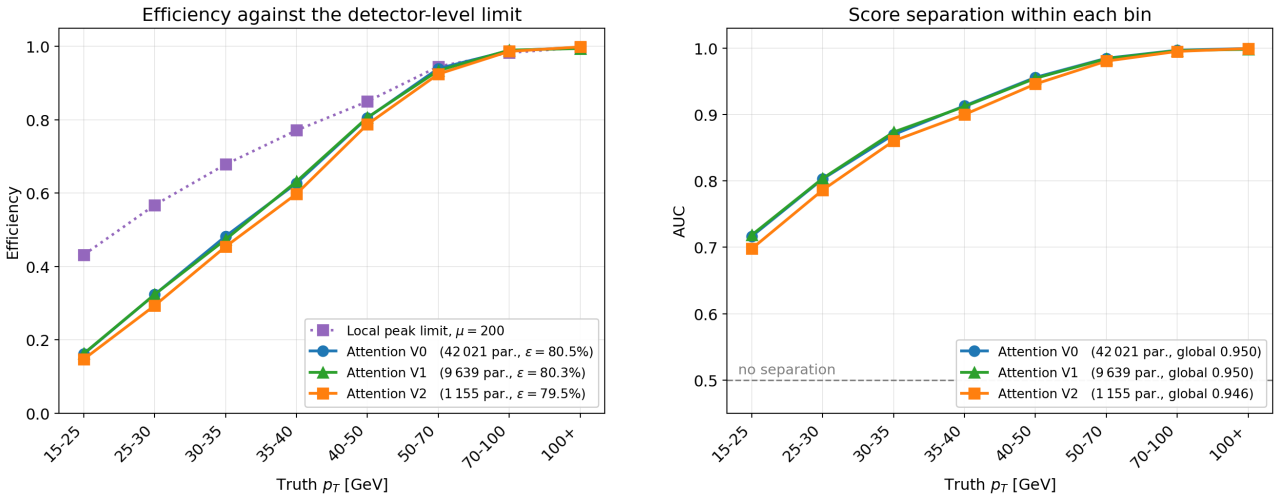


Figure 10: Performance of the Attention reference and the two architecture controls as a function of truth p_T. Left: reconstruction efficiency, including the local-peak reference at $V = 3\%$. Right: objectness AUC.

8.3 Bandwidth and FPGA considerations

The bandwidth estimate of Sec. 2.3 gives an available budget of approximately 20 Gb/s per board, corresponding to about 766 32-bit values per event. Table 5 compares this limit with the internal and output representations of the reconstruction models. Only the final jet list, and a single dense spatial map, fall within the budget; every multi-channel representation exceeds it by at least a factor of four.

Table 5: Bandwidth required by different event representations, assuming 32 bits per value.

Representation	**Shape**	**Values/event**	**Bits/event**	**Rate/board**
Calorimeter input	$50 \times 64 \times 6$	19 200	614 400	501.6 Gb/s
32-channel feature map	$25 \times 32 \times 32$	25 600	819 200	668.7 Gb/s
4-channel feature map	$25 \times 32 \times 4$	3 200	102 400	83.6 Gb/s
Single spatial map	25×32	800	25 600	20.9 Gb/s
Final jet list	20×4	80	2 560	2.09 Gb/s
20 Gb/s budget	–	~ 766	$\sim 24\,500$	20.0 Gb/s

Bandwidth alone is not sufficient to establish FPGA feasibility. The actual resource usage and latency depend on the numerical precision, degree of parallelism, reuse factor and target device, and can only be determined after synthesis. A further step is required before this synthesis can be performed: the present architectures were developed in TensorFlow, and several of their operations are not directly handled by the standard `hls4ml` Keras conversion flow. The convolutional and dense parts, together with the pooling and residual additions, are supported layer types, although their final DSP, LUT and memory cost still depends on the chosen precision and parallelism. Candidate selection and coordinate processing, which include local maxima, `TopK`, `gather` and masking as well as ϕ wrapping, $\log(1 + p_T)$ and trigonometric operations, are implemented inside custom or `Lambda` operations and therefore require rewriting or custom HLS logic. The same applies to the self-attention and normalisation stage, where `MultiHeadAttention` with residual connections and `LayerNormalization` would need to be reformulated for HLS and validated in synthesis. Finally, the detector-specific `EtaPhiPadding`

and `TowerEtaPhiLayer` are custom layers requiring an equivalent firmware implementation or a dedicated `hls4ml` extension.

These limitations do not imply that the corresponding operations cannot be implemented on an FPGA; rather, the present TensorFlow description cannot be transferred directly to firmware without modification. The current study therefore establishes the reconstruction performance, model size and approximate bandwidth requirements, while DSP, LUT, FF and BRAM utilisation, latency and initiation interval remain to be determined through a dedicated HLS implementation and synthesis on the target FPGA.

9 Conclusion

This work studied whether calorimeter information can be converted directly into a compact jet representation compatible with the constraints of the ATLAS Level-0 Global Trigger. The results for all models tested are collected in Table 3.

A consistent feature across the models is that candidate formation is considerably easier than candidate selection. Around 95% of the truth jets already have a nearby candidate before the objectness requirement, while a larger fraction is lost when signal candidates must be separated from background activity. The difficulty is concentrated at low p_T, whereas energetic jets are reconstructed with high efficiency.

9.1 What limits the reconstruction

The large improvement obtained at $\mu = 0$ shows that the performance at $\mu = 200$ is not primarily limited by insufficient model capacity. The same reconstruction strategy becomes close to fully efficient once the additional interactions are removed, while increasing the architectural complexity at $\mu = 200$ produces only a comparatively small gain. The dominant limitation is therefore associated with the information content of the calorimeter response under high pile-up rather than with the ability of the network to fit the reconstruction problem.

At low p_T, the truth jets that are rejected and the background candidates that are selected can have very similar energies and shower profiles. If the two cases look almost the same in the input, a larger or more complex network may not be able to separate them much better. This is especially important with pile-up, because the calorimeter combines energy from many interactions and does not indicate which interaction produced each deposit; some "fake" candidates may therefore be real jet-like structures from secondary interactions, making them very difficult to distinguish from the jets of the main event using only the current calorimeter information.

The $\mu = 0$ result also shows that pile-up is not the complete explanation. In the lowest-p_T interval, candidate acceptance remains above 95% while the final efficiency is only about 62%. This residual gap is important because it persists after the dominant source of background has been removed, and it suggests that part of the ambiguity is intrinsic to the present calorimeter representation or to the reconstruction objective itself.

9.2 Architectural choices

The most relevant architectural change was the introduction of candidate-level comparison. In the baseline, the objectness score is obtained locally and therefore has limited information to distinguish a weak jet from a pile-up fluctuation of similar energy; self-attention allows the score

of each candidate to depend on the other candidates in the same event and improves both the AUC and the final selection efficiency.

The control models also show that a large convolutional representation is not required to recover most of this improvement. Attention V1 reduces the model from 42 021 to 9 639 parameters while retaining almost the same efficiency and AUC as Attention V0, whereas reducing the attention block itself produces a larger loss, particularly at low p_T. Candidate comparison is therefore more important for discrimination, while the wider convolutional representation mainly benefits candidate formation and p_T reconstruction.

An important side effect of candidate-level attention is that the reconstructed jet multiplicity becomes more concentrated around the most common values in the training sample. Although the Attention model improves efficiency, purity and candidate discrimination, the correlation between predicted and truth multiplicity decreases from 0.543 to 0.512, and the survival probability of the last truth-associated candidate improves for events with three to five jets but decreases by about 17 percentage points for seven-jet events. This suggests that the attention may be learning not only which candidates look more jet-like, but also a prior on how many jets an event is expected to contain.

9.3 Bandwidth and remaining work

The final jet representation satisfies the assumed bandwidth constraint. The 20×4 jet list contains only 80 values per event and corresponds to approximately 2.09 Gb/s per board, compared with the available 20 Gb/s. The remaining question is whether the reconstruction can also satisfy the FPGA resource and latency constraints. Candidate selection, coordinate operations, normalisation and self-attention are not yet implemented in the required HLS form, so DSP, LUT, FF and BRAM utilisation, latency and initiation interval remain to be established through synthesis on the target hardware.

10 Next steps

The next steps should determine which limitations come from the available detector information and which can still be improved through the training strategy, event selection and model architecture.

Understand what the background candidates are. A first step is to identify the physical origin of the background candidates that survive the score cut, which may correspond to pile-up fluctuations, real jets from secondary interactions or other calorimeter structures. Comparing these candidates with rejected truth jets using energy, lateral spread and depth profiles would show whether there are still useful differences that the current representation does not exploit, and generator-level information could be used to determine whether some of the "fakes" are in fact real jets from pile-up interactions. A related question is whether the difficult truth jets are reconstructible from the calorimeter at all: since the current targets are particle-level jets and independent of the detector response, running a standard offline reconstruction on the same events and measuring its efficiency for the same low-p_T truth jets would provide a useful reference. If the offline reconstruction also fails for these jets, the remaining limitation is likely physical rather than specific to the trigger model.

Use event-level selection instead of a global threshold. The results show that candidate importance depends strongly on the event, yet a single global objectness threshold is currently applied. In the baseline, 1 971 rejected truth-associated candidates are still ranked within the truth multiplicity of their event, and the survival of the last truth-associated candidate



decreases strongly as multiplicity increases, so the relative ordering of candidates may contain useful information that an absolute threshold discards. Several strategies could be tested: predicting the expected jet multiplicity $\hat{N}$ and retaining the top-$\hat{N}$ candidates, using a ranking loss that compares candidates within the same event, or including an event token that learns an event-dependent score correction. The operating point should also be chosen from the trigger requirement rather than from the balanced efficiency–purity point. This is particularly clear at $\mu = 0$, where the low-p_T AUC reaches 0.977 while the final efficiency remains much lower than the candidate acceptance: the ranking is already very good, and part of the remaining loss may come from the position of the global threshold rather than from a failure to separate the candidates.

Redesign the loss and matching. The training objective should be studied again with the final architecture. The reconstruction currently uses a one-sided Chamfer loss while objectness is trained separately with binary classification, and the contribution of the p_T, position and classification terms should be measured during training to check whether one component dominates the optimisation; in particular, the normalisation between energy and position errors should be reviewed so that improvements in p_T do not hide poorer spatial reconstruction. Training and evaluation should also use more consistent matching definitions, since the regression loss, objectness labels and final Hungarian evaluation currently define associations differently. A one-to-one Hungarian assignment could instead be included during training, together with a cost for unmatched predictions, so that duplicate candidates are explicitly discouraged.

Optimise the architecture jointly. Architecture parameters should be optimised together once the final loss and model structure are defined, since individual tests can hide dependencies between layers: increasing the number of filters in a later convolution may have little effect if useful information has already been removed by the first compression layer. A systematic Optuna study could vary the convolution widths, kernel sizes, number of proposal features, attention dimension, number of heads and candidate-pool size simultaneously, with the goal of finding the smallest architecture that preserves the information needed for reconstruction and candidate comparison.

Add information beyond the current calorimeter image. Training with a controlled mixture of $\mu = 0$ and $\mu = 200$ events is worth revisiting after the loss and score calibration are improved, the hypothesis being that seeing the same physics process under very different pile-up conditions may encourage the model to rely less on the absolute energy scale and more on properties characteristic of the jet itself. If shower-shape studies still fail to separate rejected truth jets from selected pile-up candidates, however, further model complexity may provide little improvement because the required information may not exist in the current input. Timing is therefore an important direction to study separately, since it indicates when the energy was deposited and could provide a way to associate calorimeter activity with the hard-scatter interaction when the spatial and energy signatures alone are almost indistinguishable.

Move towards an FPGA implementation and test generalisation. For hardware studies, Attention V1 remains an interesting starting point because it retains almost the same discrimination performance as the full Attention model with substantially fewer parameters; the next steps should include quantisation, reformulation of unsupported operations and synthesis on the target FPGA to measure DSP, LUT, FF and BRAM usage, latency and initiation interval. Finally, the reconstruction should be evaluated on processes other than the $HH \to b\bar{b}b\bar{b}$ sample and on samples with controlled jet multiplicity and pile-up, to test whether the model has learned general jet properties or correlations specific to the current training distribution.

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